

RESEARCH PLAN

Significance

Precision health. A central goal of precision health is to tailor risk assessment, diagnosis, and treatment to specific biological, clinical, demographic, and environmental characteristics of individual patients for improving health (Topol 2014). Achieving this goal requires several important elements, including powerful artificial intelligence (AI) and machine learning (ML) methods to model the complex relationships between biomarkers and health outcomes at the population level (Topol 2019). Furthermore, methods are needed to interpret computational models and generate biological and clinical knowledge to develop new prevention and treatment strategies (Linardatos et al. 2021). The goal of this project is to develop *accessible, open, and user-friendly* AI methods and software to *automate* ML (AutoML) for analyzing multimodal data and predicting susceptibility to cardiovascular diseases (CVD). Our proposed CardioAI methodology will leverage biological knowledge to guide the selection of variables or features as part of the automated model building process. We propose here the development of CardioKB, a knowledgebase that integrates information on genes, drugs, pathways, and diseases. Additionally, our proposed CardioGraph method and software will extract high-priority features for analysis by CardioAI from the CardioKB knowledge graph. **A central goal of this project is to develop, evaluate, and deliver CardioKB, CardioGraph, and CardioAI methods with open-source products for the precision health and CVD research communities.**

The promise of AI and the rigor of prior research. As Ted Shortliffe reviews (Shortliffe 2019), AI is now more than 50 years old with renewed enthusiasm based on the successes of much publicized efforts such as AlphaGo (Silver et al. 2016), AlphaFold (Senior et al. 2020), generative AI, and chatbots using large language models such as GPT. In this context, AI refers to computational methods, software, and technology that analyze complex data in a human-like manner. That is, the goal is to create a computational system that can perform data analysis using ML, learn from that experience, and then automatically select optimal algorithms and parameter settings to perform a new analysis. This is a promising new area called automated machine learning or AutoML that was introduced less than 10 years ago. We were one of the very first groups to develop and release an AutoML algorithm and open-source software package called the Tree-Based Pipeline Optimization Tool or TPOT (A Romero et al. 2022; Le et al. 2020; Manduchi et al. 2020, 2022b; Olson et al. 2016b; a; Olson and Moore 2019; Romano et al. 2021; Sohn et al. 2017) that automatically builds an entire ML pipeline using pre-processors, feature transformers, feature selectors, and ML algorithms from the scikit-learn library (Pedregosa et al., 2011). The TPOT software is programmed in Python and available on GitHub with more than 100 current contributors from around the world. It was recently ranked as one of the top AutoML methods by KD Nuggets having received more stars on Github than other popular AutoML methods (e.g. Auto-Sklearn & Optuna). We have extended TPOT to include pipelines represented as directed acyclic graphs (TPOT2 on Github), a dataset selector operator to scale pipeline discovery to big data (Le et al. 2020), deep neural networks (Romano et al. 2021), and operators to allow the analysis of text and image data (Kumar et al., in press 2023).

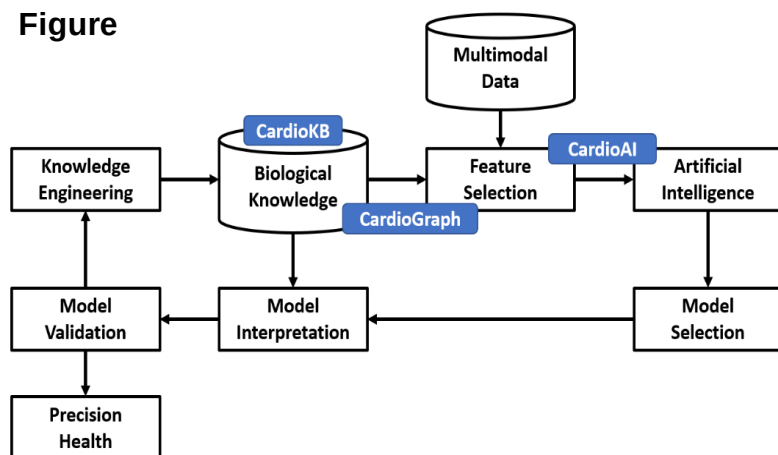
Our work on AutoML, and the work of others in this space (Feurer et al. 2015; Kotthoff et al. 2017), is important because it is difficult to know which ML methods to choose for any particular dataset. We have recently demonstrated this with more than 150 publicly available data sets where every ML method we evaluated was optimal for one or more of the datasets studied (Olson et al. 2017a, 2018a). Using our AutoML experience with TPOT, we developed Aliro to democratize ML technology to everyone through a user-friendly interface, transparent code, and powerful AI algorithms (La Cava et al. 2019; Olson et al. 2018b). The initial Aliro software has been released as open-source on GitHub along with a thorough evaluation and application to the analysis of EHR data (La Cava et al. 2020). This will serve as the basis for the new algorithms and methods in this proposal. Specifically, as shown in Figure 1 below, we will extend Aliro to include the ability to probabilistically select multimodal biomarkers for analysis using biological knowledge from **CardioKB** processed for feature selection using **CardioGraph**. These AI technologies and innovative methods for incorporating biological knowledge will enable the precision health and CVD research communities to embrace the complexity of biomarker patterns as they relate to CVD-related health outcomes. **Our motivation comes from the idea that everyone, regardless of training or expertise, should have access to user-friendly AI software tailored to precision health.** These methods can be applied to any population-based or clinical CVD study and, with minor updates to the knowledgebase, extended to other diseases and phenotypes.

The importance of knowledge engineering. AI is most effective in medicine when it has access to knowledge relevant to the problem being solved. The importance of extracting human knowledge and representing it in knowledge databases or knowledgebases has been recognized since the early days of AI in medicine (Duda and Shortliffe 1983; Feigenbaum 1984). Fortunately, bioinformatics has provided many rich knowledgebases derived from experimental results and the scientific literature. Two of the most widely used knowledgebases are the Kyoto Encyclopedia of Gene and Genomics or KEGG (Kanehisa et al. 2021; Kanehisa and Goto 2000) and Reactome (Fabregat et al. 2018). Particularly important for using AI for precision health is the integration of available knowledgebases to inform AI analyses. We propose here to develop CardioKB as a comprehensive knowledgebase relevant to CVD. CardioKB will build on our own CompToxAI knowledgebase (Romano et al. 2022) and AlzKB AI (JMIR preprint 46777, in press) inspired by HetioNet which integrates 29 different knowledge sources including KEGG and DrugBank (Himmelstein et al. 2017; Himmelstein and Baranzini 2015). We will extend and update the CompToxAI and AlzKB resources to create CardioKB by integrating new sources of knowledge relevant to CVD. For example, diet plays a very important role in risk of CVD (Chareonrungrueangchai et al. 2020) but is not captured in most knowledgebases such as KEGG. As such, we will integrate the Food Database (FoodDB) which curates more than 24,000 food-based chemicals (Naveja et al. 2018). Our CardioKB knowledgebase will be used in conjunction with new graph-based methods to inform the selection of biomarkers most likely to be related to CVD outcomes (Figure 1). These methods can be used with any CVD study.

The importance of feature selection. One approach to AI analysis is to provide all the data to the algorithm with the hope that it will be able to find the relevant patterns. While this can lead to unexpected results, there are some important limitations to this approach. First, it may not be computationally feasible for the algorithm to process the big data given the available high-performance computing resources. Second, there is an energy

cost to performing computation and thus analyzing all the data may have an undesirable environmental impact (Schwartz et al. 2020). Third, models based on big data may be harder to interpret. Feature selection addresses all of these challenges by reducing the number of variables or features using expert knowledge derived from knowledgebases or prior analytical results (Saeys et al. 2007). In addition to addressing the challenges outlined above, feature selection based on biological knowledge has the added benefit of making interpretation easier because the features used in the analysis all have a biological basis for being selected. This allows the results to be tied

Figure



directly to the knowledge and the literature from which it was derived. Indeed, we have previously shown how biology-based feature selection can benefit AutoML analysis and biological interpretation (Le et al. 2020). We will develop a CardioGraph method and software to be used in conjunction with CardioAI to use biological knowledge from CardioKB to select features for AI analysis (Figure 1). These methods can be applied to any CVD study.

Innovation

Artificial intelligence innovation. The AI methods proposed here are highly innovative as we are one of the pioneering groups in the world to design, develop, and release an AI solution to AutoML for big data. Our TPOT method and software, described above, was released in 2016 and was at least two years ahead of Google's AutoML solution and was the first to automate the building of an entire ML analysis pipeline from feature selection, feature transformation, feature engineering, ML, and parameter tuning. We have recently applied TPOT to the analysis of metabolomics data for cardiovascular disease including coronary artery disease and heart failure (Chirinos et al. 2020; Orlenko et al. 2020; Vidula et al. 2021). We are also the first to build a true AI algorithm for AutoML. Our Aliro software has a document store database that keeps a record of every ML analysis ever run on the system with an analytics engine that uses meta-features of the data and singular value decomposition to determine the best ML algorithms and parameter settings to run on any given dataset (La Cava

et al. 2020). Aliro thus learns as it experiences more and more ML results. The analytics knowledgebase that Aliro curates can be shared and used to seed other instances of Aliro. This is highly innovative and, to our knowledge, has not been done before. This highly innovative method will serve as the foundation for the proposed CardioAI algorithm and software. We will further extend this innovation by developing algorithms for feature selection that utilize biological knowledge from CardioKB and CardioGraph.

Knowledgebase innovation. Access to relevant knowledge is essential for AI research. We propose the creation of CardioKB, a knowledgebase that will integrate biomedical knowledge available in public repositories. CardioKB will be a CVD extension of our own ComptoxAI knowledgebase (Romano et al. 2022) and AlzKB AI (JMIR preprint 46777, under review) inspired by HetioNet knowledgebase (Himmelstein et al. 2017; Himmelstein and Baranzini 2015) developed by two of Dr. Moore's (PI) former students (Drs. Daniel Himmelstein & Casey Greene) with extensions by Dr. Moore's postdoc Dr. Joe Romano. Hetionet is a Neo4J graph database (i.e. knowledge graph) which consists of over 47,000 nodes of 11 types (see Figure 1) and more than two million relationships (see Figure 2) of 24 different types. Knowledge was integrated from 29 public resources to connect compounds, diseases, genes, anatomies, pathways, biological processes, molecular functions, cellular components, pharmacologic classes, side effects, and symptoms. This is the most comprehensive knowledgebase of its type in the public domain. We extended it to ComptoxAI and AlzKB AI by including a biomedical ontology to add semantic relationships among the entities. As such, it is ideal to form the basis of CardioKB. We propose to develop CardioKB by adding numerous new sources of knowledge to extend the graph. These sources will include existing knowledgebases such as the Food Database (FooDB) which curates more than 24,000 food-based chemicals (Naveja et al. 2018). CardioKB will be the most comprehensive CVD knowledgebase to date and will provide a rich source of relationships between biomarkers to inform CardioAI. To our knowledge, no other cardiovascular knowledgebases integrate such a diverse range of knowledge sources with semantic relationships provided by a biomedical ontology.

Knowledge graph innovation. Another key innovation is to connect CardioKB and CardioAI through development of novel Cardiovascular Graph (CardioGraph) algorithms. We will develop a novel knowledge-driven feature selection approach (Aim 2.1) and a data- and knowledge-driven feature selection approach (Aim 2.2) to provide importance weights to multi-modal biomarkers in CardioAI (Aim 3). Firstly, our proposed network propagation framework represents a significant departure from the current norm by enabling the integration of multiple layers of biological networks and incorporating known directionality of interactions into a coherent framework. It also allows the integration of prior biological knowledge. Using a random walk with restart approach, we can identify and prioritize novel biological elements associated with multimodal biomarkers in a network with multi-layer information structures, their relationship, and the direction of interactions. Our network propagation approach constructs an explicitly 'heterogeneous' network, considering different node and edge types. Leveraging heterogeneity, the metapath-based random walk selectively evaluates biologically relevant paths. This causality-aware network propagation method aids in identifying and prioritizing multi-modal biomarkers for CardioAI. Secondly, as a complementary approach, we will develop a novel network-based algorithm to prioritize the selection of multi-modal cardiovascular-related biomarkers and omics features by combining data- and knowledge-driven graphs from CardioKB. This highly innovative method will not only integrate multi-modal datasets but also leverage CardioKB to assign importance weights to biomarkers, serving as prior knowledge for the probabilistic feature selector of CardioAI in Aim 3. CardioGraph acts as an intermediary algorithm that extracts a key subset of omics features from heterogeneous multi-modal datasets based on data and knowledge, making it easily accessible to novice users for new analyses.

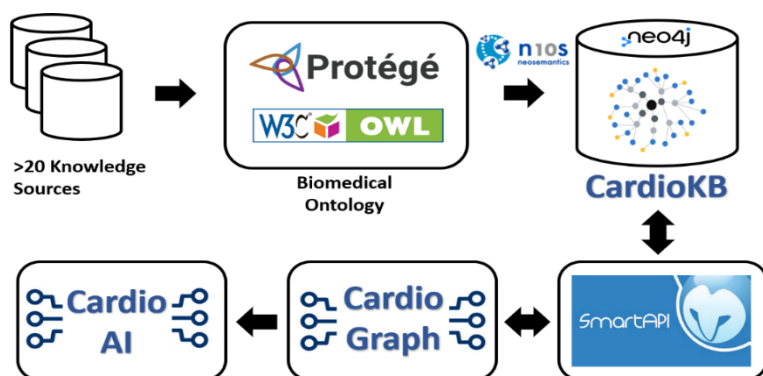
Approach

We describe below our approach for developing a CVD knowledgebase (**CardioKB**) (Aim 1), a **CardioGraph** algorithm for identifying biomarkers based on their biological relationships (Aim 2), and a **CardioAI** algorithm and software for AutoML using biological knowledge from CardioKB and CardioGraph (Aim 3). We then describe our data analysis plan in Aim 4 and our plan to distribute open-source software in Aim 5. *Note all software developed by this program will be available to anyone interested in using ML to carry out multiomic analysis of CVD regardless of data source or study.*

Preliminary Studies – Dr. Moore (PI) has extensive experience applying ML algorithms to detecting clinical and genomic biomarkers associated with CVD. Better risk stratification strategies are needed to enhance clinical care

As we have done previously, we will construct the ontology manually using the Protégé ontology editor (v.5.5.0), following an iterative process. First, we will prototype a class hierarchy containing the types of nodes (e.g., genes,

Figure 3



drugs, pathways, and diseases) desired in the knowledge base. We will then annotate these classes with data properties (node properties) and object properties (relationship types). Finally, we will place restrictions on the ontology to reflect biology. For example, there are restrictions stating that all pathways contain one or more genes, or that all drugs in the knowledgebase must have a valid DrugBank ID. We will repeat these steps several times, making revisions on previous iterations until the semantic ontology contents are consistent with current knowledge as we have done previously for CompToxAI and

AlzKB AI. After obtaining a copy of the ontology populated with entities, relationships, and data properties drawn from third-party data sources, we will serialize the ontology into the RDF/XML graph data format. We will then populate a Neo4j graph database with the contents of the RDF/XML file using the Neosemantics library from Neo4J. Finally, we will strip the newly populated graph database of unnecessary artifacts that are components of the OWL 2 standard, leaving only nodes, relationships, and node/relationship properties defined within the hierarchy. For the publicly hosted version of CardioKB, we will create a web server that hosts both the static website (containing information, documentation, and usage details) and the Neo4j graph database available by navigating to a subdomain of the main website as we have done for CompToxAI and AlzKB AI. Users will be able to query the graph database using Cypher and our own user-friendly search tool that we will tailor specifically to CVD.

We propose to create CardioKB by adding the following sources of knowledge to extend the graph. First, we plan to extract knowledge about metabolites and their connections with genes and diseases. We will utilize the Human Metabolome Database (Wishart et al. 2018) or HMDB to link over 100,000 water and lipid soluble metabolites to elements already in the knowledgebase such as genes (e.g. GenBank), pathways (e.g. KEGG) and drugs (e.g. DrugBank). These metabolites will be linked to the PhytoHub database containing information about 1,200 polyphenols, terpenoids, alkaloids and other plant secondary metabolites present in commonly consumed foods. The metabolites will also be linked to the Phenol-Explorer database (Rothwell et al. 2013) which contains 35,000 content values for 500 different polyphenols in over 400 foods as well as the Exposome-Explorer (Neveu et al. 2017) which links more than 145 dietary biomarkers with environmental exposures. In addition to metabolites, nutrients, and food-based chemicals, we will also link this knowledge with gut microbiome using the Virtual Metabolic Human database (Noronha et al. 2019). Finally, we will link all of these entities, and others already in the knowledgebase such as compounds, to the Food Database (FoodDB) which curates more than 24,000 food-based chemicals (Naveja et al. 2018). We will also tailor the diseases in CardioKB to cardiovascular (e.g. coronary artery disease, heart failure, etc.) and cardiovascular-related diseases (e.g. type 2 diabetes, essential hypertension, etc). These new sources of knowledge will transform CompToxAI & AlzKB AI to CardioKB which then be used by CardioGraph to prioritize features (i.e., biomarkers) for ML analysis using CardioAI (Figure 3). These are described below.

The **potential challenges of CardioKB** are minimal given we have already developed CompToxAI and AlzKB AI using the outlined biomedical ontology and Neo4J graph database. Both developed resources have been publicly deployed and CompToxAI has already been peer-reviewed and published (Romano et al. 2022) along with use for knowledge-guided deep learning (Hao et al. 2022). Potential pitfalls include the incorporation of new knowledge sources such as those related to metabolites, foods, and nutrients as outlined above. These will need to be mapped to the ontology which could generate alignment problems and lack of granularity. Further, continuous maintenance of the resource can also be a challenge. However, we have solved all alignment and granularity problems for more than 20 different knowledge sources without fail. Our team is experienced in building these knowledgebases and working with the ontology. Also, we have dedicated staff updating the existing resources and they will the do the same here. It is worth noting that the ALzKB graph contains 118902 nodes and 1309527 edges. This is computable. For example, it took about an hour to run 15 different community

detection algorithms (~ 5 minutes each) on the entire graph. The proposed CardioKB graph will be bigger, but not significantly so.

Methods for Aim 2 – Develop a Cardiovascular Graph (CardioGraph) algorithm: The aim of this section is to develop and evaluate algorithms that prioritize the selection of multi-modal cardiovascular biomarkers by utilizing measures of network topology applied to the knowledge graph obtained from CardioKB. We propose the development of complementary CardioGraph algorithms, including knowledge-driven feature selection (Aim 2.1) and data- and knowledge-driven feature selection approach (Aim 2.2), which will provide importance or weights to multi-modal biomarkers to CardioAI in Aim 3.

Aim 2.1 Knowledge-driven feature selection. We will develop a network propagation algorithm to prioritize and extract multi-modal biomarkers from the CardioKB knowledge graph. This algorithm will provide cardiovascular-specific expert knowledge to CardioAI in Aim 3. Our proposed approach consists of the following steps: (1) determination of biologically meaningful metapaths and (2) parameter tuning using a cross-validation approach.

Metapath-based random walks. The directionality of biological interactions can be leveraged by constraining the propagation process with a reverse direction. For instance, only the upstream paths (e.g., from a reaction to the corresponding enzyme) will be used to prioritize multi-modal biomarkers in the upstream direction. The constraint for network propagation paths can act not only on reverse directions but also on the biological relevance of such paths. We will compare two strategies (1) using only the reverse directions and (2) using a set of biologically meaningful paths. A metapath serves as a ‘template’ for specifying the types of nodes and edges in an actual path. We will then simulate the propagation using only the paths that fit these templates. The metapath approach has been shown to improve the performance of many graph-mining tasks as it restricts the vast space of all possible paths into a smaller and relevant space. We will compare both “generous” (with reverse constraint only) and “restrictive” (with biologically meaningful metapaths) approaches. This procedure will effectively incorporate the relationships embedded in the biological databases into our network propagation method. Biologically meaningful metapaths will be obtained by combining fundamental biological processes and common disease pathways.

Given these metapath constraints, our propagation algorithm will use a random walk with restart (RWR) approach. A random walk is a path generated by choosing a random neighbor at each step, starting from a given seed. Namely, P_i^e , the relevance of node i given the seed e , can be written as

$$P_i^e = \alpha \sum_j \frac{A_{ij}}{k_j} P_j^e + (1 - \alpha) \delta_i^e,$$

where α is a tuning parameter, A_{ij} is the adjacency matrix, k_j is the degree of node j , and δ_i^e is the Kronecker’s delta. In the vector and matrix form,

$$\vec{P}^e = (1 - \alpha)(\mathbf{I} - \alpha \mathbf{A} \mathbf{K}^{-1})^{-1} \vec{\delta}^e.$$

By setting random initial values and iteratively applying this formula (known as the “power method”), we can calculate the relevance score of each node in the network. For example, if two genes are strongly associated through many paths in the network, a random walk starting from one of the genes is likely to reach the other gene quickly. We will employ a popular implementation called “random walk with restart” (RWR) or “personalized PageRank”. In RWR, random walks return to the seed nodes and restart the process with a given probability. This procedure enables us to quantify the proximity of each node to our seeds, which we interpret as relevance to the disease. Genes that are well connected, both directly and indirectly, to our seeds will have higher scores, while genes that are distant from other disease genes and relevant metabolites will have low scores. In addition, we will also utilize the degree-neutralized egocentric PageRank (DNEP), which incorporates a term to neutralize the effect of degree:

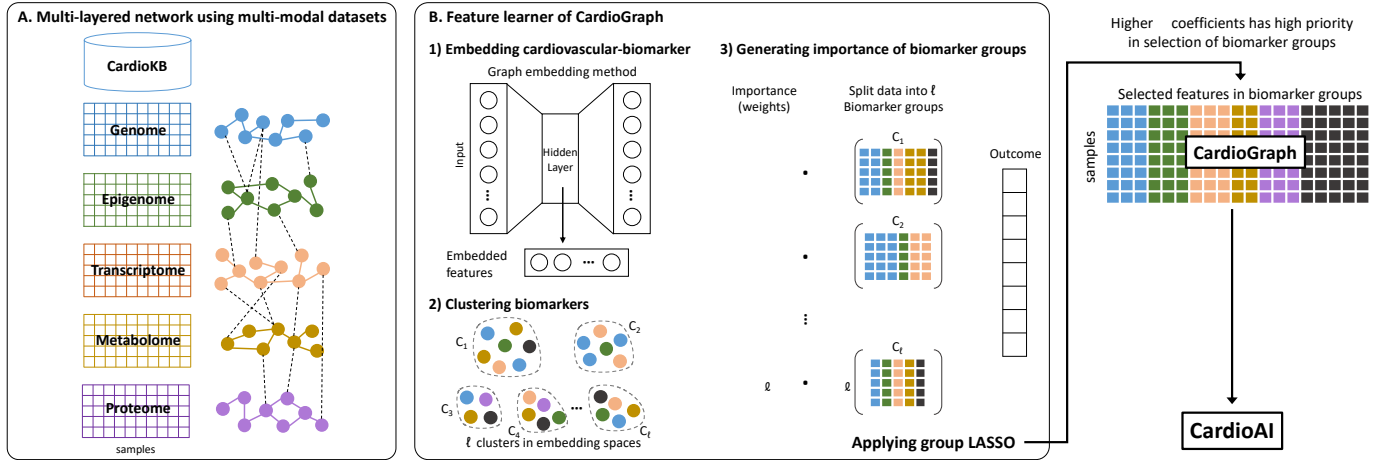
$$D_i^e = \alpha k_i^{-1} \sum_j A_{ij} H_j D_j^e + (1 - \alpha) \delta_i^e,$$

where $H_j = [\sum_l A_{lj} k_l^{-1}]^{-1}$. We can then combine the original personalized PageRank with the degree-neutralized PageRank to obtain the relevance score $R_i^e(\beta) = (1 - \beta)P_i^e + \beta D_i^e$, which has been shown to produce highly relevant neighborhood recommendations.

Cross-validation and parameter tuning. Once we obtain the heterogeneous network, we will employ cross-validation techniques using known genetic factors to tune and select the optimal model. Our model will include hyperparameters related to network construction (e.g., determining edge weights for edges appearing in multiple databases), network seeding (e.g., selecting appropriate threshold values for network seeds), and random walk (e.g., return probability). These parameters can be fine-tuned through cross-validation. In other words, we will leave out one known genetic factor at a time and evaluate the model's performance across the parameter space. We will consider the correct identification of known genetic factors for human disease as a proxy of the true model performance. Additionally, we will conduct a sensitivity analysis on the network structure by bootstrapping the edges and repeating the simulation on the ensembles.

Aim 2.2 Data- and knowledge-driven feature selection. We will develop a complementary network-based algorithm to prioritize the selection of multi-modal cardiovascular biomarkers by combining data- and knowledge-driven graphs from CardioKB. The primary objective of Aim 2.2 is to prioritize heterogeneous biomarker groups by integrating multi-modal datasets and CardioKB. We aim to provide importance or weights of biomarkers, serving as priors for a probabilistic feature selector of CardioAI in Aim 3 (see Figure 4).

Figure 4



Constructing data- and knowledge-driven multi-layered networks. We will construct a multi-layered network using multi-modal datasets and CardioKB (from Aim 1). A multi-modal biomarker network can be represented as a multi-layered graph, $G(V, W, L)$, where node V denotes set of biomarkers, W denotes similarity between nodes, and L denotes the collection of K independent modality pre-defined from CardioKB, $L = \{L_1, L_2, \dots, L_K\}$ (Kivelä et al. 2014). Figure 4 illustrates an exemplary multi-layered multimodal biomarker network with six independent modalities such as genome (L_1), epigenome (L_2), transcriptome (L_3), metabolome (L_4), proteome (L_5), and phenome (L_6). The total number of biomarkers (nodes) is $N = n_1 + n_2 + \dots + n_k$, where n_k stands for the number of biomarkers in K^{th} modality. The similarity matrix W is an $N \times N$ block matrix can be decomposed into intra-modality (block diagonal matrix) and inter-modality (off-diagonal block matrix). The edge weights in intra-modality w_{ij} between two biomarkers v_i and v_j is obtained from data-driven sources, calculated by Gaussian similarity kernel, $w_{ij} = e^{(-\|v_i - v_j\|_2^2 / \sigma^2)}$. In the inter-modality, the edge values between different modalities can take the value 1 if there is a knowledge-driven association between two biomarkers in different layers; otherwise, it is set to 0. Thus, each $G(V, W, L)$ consists of data-driven relations within a single modality and knowledge-driven relations between different modalities.

Feature learner of CardioGraph for prioritizing a set of biomarkers. We will generate a set of biomarkers to reduce computational complexity for feature selection of CardioAI and will prioritize biomarker groups based on their importance. The algorithms primarily consist of three parts: a) embedding cardiovascular biomarkers, b) clustering biomarkers, and c) generating biomarker importance.

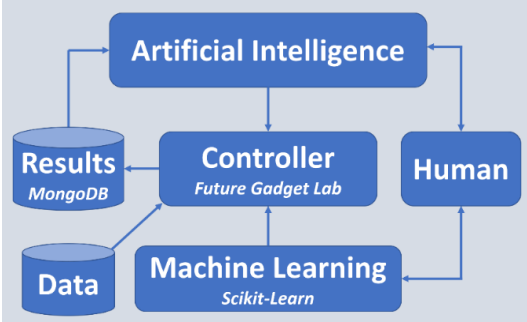
- a) Embedding cardiovascular biomarkers: We will perform biomarker embedding to capture characteristics of the multi-layered multimodal biomarker network. We will apply the node2vec algorithm to find embeddings of biomarkers in low-dimensions while preserving the original network topology. Node2vec can learn low-dimensional representations for biomarkers in a graph by optimizing a neighborhood-preserving objective. Given a biomarker $v \in V$ from the multi-layered multimodal network $G(V, W, L)$, we can define nearby biomarkers $N_s(v)$ with strategy S (Goyal and Ferrara 2018). Then, we can obtain low-dimensional embedded biomarkers f by learning $f: v \rightarrow \mathbb{R}^d$ with the following objective function, $\max_f \sum_{v \in V} \log \Pr(N_s(v) | f(v))$. As a result, the embedded biomarker f contains both data- and knowledge-driven information.
- b) Clustering biomarkers: Using the embedded biomarkers, we will perform a clustering algorithm such as k -means spectral clustering to group a set of heterogeneous biomarkers (Ng et al. 2001). Heterogeneous biomarker groups are sets of cardiovascular-related biomarkers that exhibit similar patterns in embedding spaces and belong to various heterogeneous modalities. Let the set of biomarker groups be $F = \{F^1, \dots, F^\ell\}$, where ℓ is the number of clusters and the total number of biomarkers is $|F| = N$. In Figure 4, group 1 has 7 heterogeneous biomarkers, including 2 genomes, 1 epigenome, 1 transcriptome, 2 metabolome, and 1 phenome. The list of heterogeneous biomarker groups will serve as a pre-defined feature set for CardioAI's sophisticated feature selection in Aim 3.
- c) Generating importance: The importance of biomarker groups will be determined by identifying associations between ℓ heterogeneous biomarker groups and coronary heart disease outcome y . Based on their importance, the list of biomarkers/omics features in a specific group can be used as simple feature selection in building predictive models or as prior probabilities of more sophisticated and precise feature selection for CardioAI in Aim 3. We will apply group LASSO to calculate the importance or weights for ℓ heterogeneous biomarker/omics groups (Meier et al. 2008). Let $\mathbf{X}$ denote the collection of n samples, $F = \{F^1, \dots, F^\ell\}$ denote the pre-defined groups according to biomarker groups, and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_\ell)$ denote a vector of coefficients with respect to pre-defined groups. Then, the input matrix $\mathbf{X}$ is decomposed of ℓ groups $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_\ell$, with p_ℓ denoting the size of group ℓ . The objective function of group LASSO becomes $\boldsymbol{\beta}^* = \operatorname{argmin} \|\mathbf{y} - \sum_{i=1}^\ell \mathbf{X}_i \beta_i\|_2^2 + \lambda \sum_{i=1}^\ell \sqrt{p_i} \|\beta_i\|_2$. The optimized weight $\boldsymbol{\beta}^*$ can serve as a priority for groups and can be used as prior knowledge for CardioAI.

The **potential challenges of CardioGraph** include computational complexity that arises when dealing with a large number of modalities (layers) or biomarkers (features). However, it is important to note that CardioGraph focuses on the embedding problem of large-scale networks, as its ultimate objective is to prioritize feature groups through multi-modal biomarker clustering in the embedding space. Fortunately, there are numerous methodologies currently being developed for embedding large-scale networks, which can practically address these potential challenges. In our previous work, we successfully constructed large-scale networks such as SNP-SNP networks and multi-layered disease-gene-drug networks (Nam et al. 2023), performed downstream analysis using these networks. Therefore, we are confident that the complexity problem can be resolved.

Methods for Aim 3 – Develop a Cardiovascular Artificial Intelligence (CardioAI) algorithm and open-source software for automated machine learning (AutoML) analysis of multimodal biomarker and health outcomes data informed by the CardioKB knowledgebase and CardioGraph feature selection. The primary goal of our AI research strategy is to bring powerful automated ML (AutoML) methods and open-source user-friendly software to the broader precision health and CVD research communities to enable everyone, regardless of computational expertise, to embrace the complexity of this common disease for biomarker discovery. Our Aliro platform (open-source on Github) represents one of the very first AutoML methods that has a memory of every analytical result such that an AI algorithm that can learn from that experience to suggest and launch new ML analyses. A major benefit of this approach is that it makes ML accessible to naïve or novice users because it takes the guesswork and complexity out of picking, running, and optimizing ML algorithms. Bringing this technology to the CVD research community will open the door to broad adoption of ML methods for embracing the complexity of the disease for biomarker discovery. We propose here a novel CardioAI platform building on Aliro by adding feature selection informed by biological knowledge. We describe here the Aliro algorithm and software and then present below our proposed MultiModal Feature Selection (M2FS) algorithm for transforming Aliro to CardioAI.

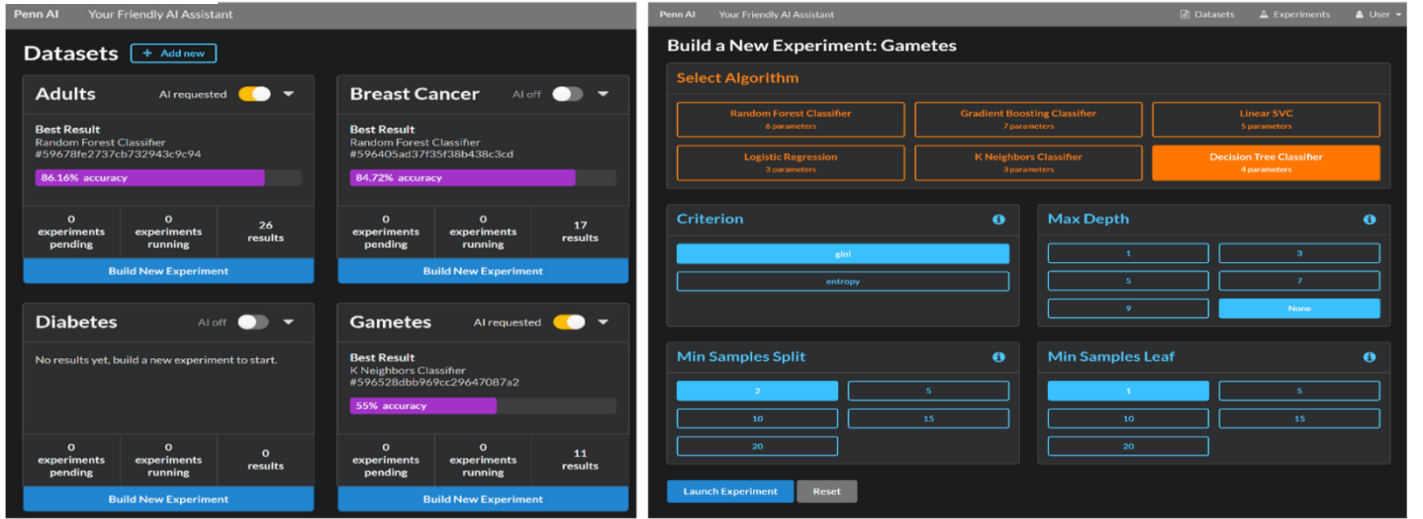
Aliro consists of several important components (see Figure 5). First, we draw on a widely-used and open-source ML library called Scikit-Learn (Pedregosa et al. 2011) that provides many of the commonly use. Second, we adopted an open-source controller called Future Gadget Lab (FGL) that allows Aliro to automatically select a ML algorithm and a dataset for analysis on a server. The FGL controller then retrieves the result, encodes it in JSON format, and deposits it in our database. Third, we selected MongoDB as a freely available database. MongoDB is an open-source document store that makes it very easy to deposit and retrieve ML results where each results file contains the details of the data and the analysis as well as the results. It is also a graph database making it possible to link results. Fourth, we built a novel AI algorithm that performs an analysis of all the results and recommends the best ML algorithm to run on a dataset based on prior performance and meta-features of the data such as sample size, distribution, and correlation structure (La Cava et al. 2020). Finally, we built a user-friendly graphic-user interface (see Figure 6 below) to allow naïve users to perform an interactive ML analysis or to call the AI for automated analysis by pushing a few buttons. The interface was built entirely in

Figure 5



JavaScript and HTML5. Aliro is designed around several Docker containers that make it easy to download and install on a server or cluster/cloud computing environment. It is designed to be used locally to avoid data and results privacy concerns. Because the interface is web-based it can be accessed over the internet and is designed to be used from a smartphone or mobile device. The open-source and user-friendly nature of Aliro will make it highly accessible to the broader precision health and CVD research communities.

Figure 6



The CardioAI recommender system. The key to CardioAI is the recommender system which selects optimal ML methods and parameter settings for a given dataset based on its history. We recently evaluated multiple different recommender algorithms and showed that a singular value decomposition (SVD) method was optimal. The SVD recommender is a collaborative filtering method popularized by the top entries to the Netflix challenge (Bennett et al. 2007). SVD is based on a matrix factorization technique that attempts to minimize the error of the rankings via stochastic gradient descent (SGD). Each rating is estimated as

$$\hat{r}_{ad} = \mu + b_a + b_d + \mathbf{q}_a^T \mathbf{p}_d.$$

Here, μ is the average score of all datasets across all learners; b_a is the estimated bias for algorithm a , initially zero; b_d is the estimated bias for dataset d , initially zero; $\mathbf{q}_a$ is a vector of factors associated with

$$\mathcal{L} = \sum_{r_{ad} \in \mathcal{D}} (r_{ad} - \hat{r}_{ad})^2 + \lambda (b_a^2 + b_d^2 + \|\mathbf{q}_a\|^2 + \|\mathbf{p}_d\|^2).$$

algorithm a and $\mathbf{p}_d$ is the vector of factors associated with dataset d , both initialized from normal distributions centered at zero. Ratings are learned to minimize the regularized loss function

A novel MultiModal Feature Selection (M2FS) algorithm. The goal of our novel MultiModal Feature Selection (M2FS) algorithm is to convert feature importance weights from CardioKB and CardioGraph to probabilities CardioAI can use to identify feature sets likely to be informative for ML modeling. A further goal is for the probabilities to be dynamic and adaptive in response to evolving ML results. To accomplish this, we propose the following M2FS algorithm. Let F be a vector of $a = 1$ to A probabilities where F_a is the probability of selecting

$$F_{a,i+1} = F_{a,i} + F_{a,i} * Q_a^\alpha * Q_a^\beta * Q_a^\gamma * Q_a^\delta$$

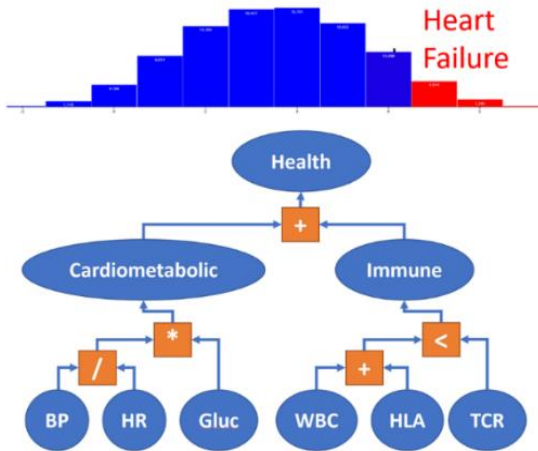
feature a from the set of all A features for inclusion in a ML model. We initialize the vector F to $1/A$ such that each pair has the same uniform probability of being selected. For $i = 1$ to n runs of CardioAI, generate n datasets with probability threshold K . For each of the 1 to n CardioAI runs, update the F probabilities according to the following function where Q is the quality of feature a , and the coefficients determine the relative weighting of Q for each source of knowledge (e.g. Q_a^α for metabolite scores, Q_a^β for transcriptomic scores, etc.). After each iteration of the algorithm, the A probabilities are scaled so they sum to unity. The advantages of this approach are its simplicity, its time complexity, and the ability to incorporate multiple modes of knowledge such as metabolite scores. This algorithm is fast because only vectors of probabilities are stored and manipulated. This is important given the large number of features that will be available from TOPMed.

Evaluation plan. There are two primary components to our evaluation plan. **First**, we describe how we will simulate realistic data. **Second**, we describe our experimental design and analysis plan.

We will simulate biomarkers and case-control endpoints with our newly developed Heuristic Identification of Biological Architectures for simulating Complex Hierarchical Interactions (HIBACHI) method and software (Moore et al. 2015, 2018b). This approach combines a biological hierarchy, a flexible mathematical framework, a liability threshold model for defining endpoints and a heuristic search strategy for identifying complex models of susceptibility. We have previously used HIBACHI to benchmark the performance of different ML methods (Moore et al. 2018b) and to understand the role of epistasis in common diseases (Moore et al. 2018a). **This approach has been validated in several different ways.** First, we demonstrated that HIBACHI models representing genetic effects were highly consistent with those observed in GWAS studies (Moore et al. 2020).

Second, we demonstrated that it could be used to discover a variety of different benchmark data models for evaluating multiple different ML models with reproducible results (Orzechowski and Moore 2022) including a new conservation ML approach that we developed (Sipper and Moore 2021). Third, we recently collaborated with Dr. Holly Mortensen from the NIH to use it for evaluating graph-based ML for assessment of genetic effects on adverse outcome pathways in the environmental sciences (Romano et al. 2023).

Figure 7. HIBACHI simulation.



of functions that can then be used with the liability threshold model described below to generate disease status. Cases and controls can then be sampled from this distribution.

At each combination of clinical measures, the user can specify one of 25 mathematical functions organized into groups labeled Basic, Logical, Bitwise, Unary, Large and Misc. For example, Basic functions include addition (ADD or +), subtraction (SUB or -), multiplication (MULT or *), division (DIV or ÷), modulus (MOD), and modulus-2 (MOD2). Logical functions include greater than (GT or >), less than (LT or <), AND (&&), OR (||), and XOR (^). These provide a comprehensive array of possible functions with 25^6 combinations. We use a liability threshold model to simulate disease from the distribution of phenotypic values generated from the variables and functions as described (see Figure 7). The user can select the liability threshold to achieve a desired disease prevalence. HIBACHI generates liability distributions with diverse interpretations. We have previously shown that

this approach can model complex hierarchical gene-gene interactions in a biological manner and that these simulated data are useful for evaluating ML algorithms (Moore et al. 2015, 2020). Once we have generated a large population of subjects (e.g. $n=100,000$ or more) we will draw from these data to evaluate the performance of M2FS.

We will then compare the performance of different parameters and different ML methods using analysis of variance (ANOVA) where the outcome is a performance measure and the predictors are features such as NGFS weights, genetic model, number of variants, method, and so forth. We will carry out post-hoc tests (e.g. Tukey's) to identify the best parameter settings (Tukey 1949). Specifically, we will design all evaluation experiments using a full factorial treatment design (Box et al. 1978). Assuming a significance level of 0.05, we estimate that we will need 10 datasets for each of the 400 parameter combinations to have more than 80% power to detect a small difference in means of 0.5 standard deviations. We will have more than 99% power to detect effect sizes of 0.75 or greater. Thus, only 4,000 total simulated datasets ($10 * 400$) will provide more than ample power to detect a small difference in performance between different implementations of M2FS. Additional simulated data can easily be added if necessary.

The **potential challenges of CardioAI** are minimal given this is an existing mature algorithm that has been open-source for at least five years with continuous feedback and improvement including peer-review. The M2FS algorithm is new and there is no guarantee it will work as expected. However, the algorithm is based on weighted probabilities and can be easily modified if performance is an issue. If we are unable to produce a high-performing M2FS algorithm we will explore adapting ReliefF algorithms using the same kinds of weights. We have a long track record of developing ReliefF algorithms (Greene et al. 2009; Hao and Moore 2021; Moore 2015; Urbanowicz et al. 2018b; a). As another option we could explore the use of weights as part of recursive feature elimination using ML methods such as XGBoost. We have previously built ML pipelines using recursive feature elimination for the analysis of GWAS data (Manduchi et al. 2022a). To our knowledge, this has never been done before and thus would be novel and could be explored as alternative approaches if there is time.

Methods for Aim 4 – Apply CardioKB, CardioGraph, and CardioAI to building predictive models of CVD outcomes using multimodal biomarkers including genetic, epigenomic, transcriptomic, metabolomic, and proteomic measures. We will apply the new methods and software to the analysis of multimodal data for the prediction of CVD and CVD-related diseases including coronary artery disease, heart failure, type 2 diabetes, and essential hypertension using data collected from the TOPMed program as described above and UK Biobank as necessary. CVD-related knowledge will be provided by CardioKB with features prioritized for analysis by CardioGraph and adaptively selected and analyzed by CardioAI and the proposed M2FS algorithm for feature selection. We describe below how we will address false-positives due to overfitting along with our strategies for ensuring reproducibility and interpretation of ML models.

False positives. A central concern with any AI or ML analysis is the detection of false-positive signals that come from the algorithm learning random patterns specific to a single data set. We will address this in several ways. **First**, CardioAI has 10-fold cross-validation built into its ML methods. This helps assess the generalizability of the results and thus reduces false positives (Coffey et al., 2004a, 2004b; Moore, 2010). **Second**, we will also evaluate the consistency of model selection across cross-validation intervals (Moore 2010). **Third**, statistical significance will be evaluated using our own permutation testing tools that take advantage of the extreme value distribution for efficiency (Pattin et al. 2009) and our own novel permutation testing method which can take into account nonadditive interactions among biomarkers while holding the additive effects constant (Greene et al., 2010a).

Reproducibility. The ability to reproduce CardioAI results is of utmost importance. **First**, CardioAI is designed to maximize the reproducibility of its results by using Docker containers (Beaulieu-Jones and Greene 2017). This allows the operating system, libraries, and exact versions of the software to be shared. **Second**, we will utilize our own proportional-instance cross-validation (PICV) algorithm that improves the reproducibility of ML results by creating more homogenous splits of the training and testing data (Piette and Moore 2018). **Third**, all models will be developed and validated using data from TOPMed and UK Biobank.

Interpretation. We will carry out both statistical and biological interpretation of the results. **First**, we will carry out a sensitivity analysis to evaluate the information provided by each feature to the overall ML result. We will use

the permutation feature importance scores method that randomizes each feature individually, reruns the ML analysis, and then measures the change in predictive accuracy or AUROC. **Second**, we will use local surrogate models to explore individual model predictions. Here, we will use the popular local interpretable model-agnostic explanations (LIME) approach for building local surrogate models (Ribeiro et al. 2016). We will also use Shapley values that have the advantage of guaranteeing that the predictions are fairly distributed among features (Shapley 1952) as we done for AutoML with CVD (Manduchi et al. 2022a). Permutation feature importance scores will be used to assess the relative contribution of each feature to the quality of the classifier. **Third**, we have developed and pioneered the use of information theory (i.e., entropy) to decompose ML models into additive, non-additive and redundant components (Fan et al. 2011; Hu et al. 2011, 2013a; b; Moore et al. 2006). All statistical relationships will be visualized using our tools for interaction dendrograms (Moore et al. 2006) and interaction networks (Hu et al. 2011, 2013c). This includes our novel ViSEN software for visualizing two-way and three-way interaction effects (Hu et al. 2013c). These post-hoc analyses will provide information about the additive (i.e., independent), non-additive (i.e., synergistic interaction), and redundant (i.e., correlation) relationships among the features in the ML models. **Fourth**, we will use our own CardioKB knowledgebase to assist with interpretation. All biomarkers identified by CardioAI can be assessed using the CardioKB knowledge graph for biological relationships. **Fifth**, we will collaborate closely with Dr. David Ouyang (Co-I), our cardiovascular domain expert, to interpret all results.

Methods for Aim 5 – Distribute and support CardioKB, CardioGraph, and CardioAI as open-source software via Github. The goal of this aim is to deliver each new software package as a product to the consortium and the broader precision health community. We will release all software as open-source through our GitHub account so everyone can benefit from the software. Both Drs. Moore and Kim have extensive experience developing, evaluating, releasing, and supporting open-source ML and AI software using best practices such as version control with Github. For example, Dr. Moore has recently released open-source ML software packages on Github including Aliro for AutoML using python and docker (La Cava et al. 2020), tree-based pipeline optimization tool (TPOT) for automated ML and its extensions to big data using python (Le et al. 2020) and a TPOT 2.0 version, evolutionary biclustering (EBIC) for unsupervised ML using python (Orzechowski et al. 2018b), row-based biclustering for unsupervised ML using R (Orzechowski et al. 2018a), the HIBACHI method and software for complex data simulation in python (Moore et al. 2018b), the TreeHeatR method for visualization and interpretation of ML methods in R (Le and Moore 2020), and the ReBATE package for feature selection using ReliefF-based algorithms in python (Urbanowicz et al. 2018b).

Preliminary studies for CardioAI. The heart of this project is CardioAI which will automate and make accessible ML for the analysis of multimodal data. We recently performed an evaluation of Aliro, the precursor to CardioAI, using over 150 real data sets (La Cava et al. 2020). We also applied Aliro to the analysis of EHR data and compared its performance to deep learning neural networks. We briefly summarize this study here to demonstrate that the platform for CardioAI development has been evaluated and peer reviewed.

We assessed each recommender on 165 open-source classification datasets, varying in size (hundreds to millions of samples) and origin (bioinformatics, economics, engineering, etc.). We used datasets from our own Penn Machine Learning Benchmark or PMLB (Olson et al. 2017b). In previous work, we conducted an extensive benchmarking experiment to characterize the performance of different machine learning frameworks on these problems (Olson et al. 2018a). The benchmark assessed 13 machine learning algorithms over a range of hyperparameters. This resulted in a cache of over one million machine learning results across a broad range of classification problems which we use here to assess the performance of each recommender with a known ranking of algorithms for each dataset. Here, we used a subset of these results consisting of 12 machine learning algorithms with 7580 possible hyperparameter configurations (La Cava et al. 2021). We determined that a singular value decomposition (SVD) recommender had the best overall performance compared to other methods such as clustering and KNN. We showed that Aliro was able to automatically select the right machine learning method and hyperparameter settings after experiencing as few as 50 different datasets out of the 165 used to evaluate it. This performance demonstrated that the AI is indeed learning from experience and doing so after seeing only several dozen datasets. SVD is the algorithm currently being used by Aliro to select machine learning algorithms and hyperparameter settings.